

Jacob(Hyunseong) Yoon

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EDUCATION

Northeastern University, Boston, MA
Bachelor of Science in Major: Electrical Engineering | Minor: Robotics

May 2028
GPA: 3.72

RELEVANT COURSES

Circuits and Signals, Embedded Design, Computing Fundamentals, Fundamentals of Networks

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB, SQL
Embedded/Hardware: STM32, Arduino, Raspberry Pi, FPGA, PWM, encoder interfacing, UART, I2C
PCB/EDA: Altium Designer, KiCad, LTSpice, Quartus Design: SolidWorks, AutoCAD
Lab/Build: Soldering, troubleshooting, AC/DC circuits, relay-based systems
Networking: Cisco/HPE switch configuration, VLANs, IP management, fiber optics, UTP cabling

EXPERIENCE

Electrical Engineer September 2025 - Present
Northeastern Mars Rover Team, Boston, MA

- Designed and fabricated PCB prototypes for temperature-sensing subsystems using KiCad
- Integrated an STM32 microcontroller over I2C to acquire sensor data and validated functionality through bench testing

Network Systems Maintenance August 2023 - May 2025
Sergeant, Republic of Korea Air Force

- Managed base-wide network infrastructure by configuring and troubleshooting Cisco/HPE switches and performing precision installation and termination of fiber-optic and Cat5e cabling.
- Administered logically separated network domains (VLANs) and IP address management (IPAM) to ensure secure, high-availability operations across multiple systems.
- Optimized network reliability by developing standardized diagnostic procedures using network testing tools to isolate physical-layer and configuration faults.
- Engineered emergency notification systems for time-sensitive alerts in base facilities, designing relay-based control circuits with AC-to-DC power conversion to drive DC buzzer outputs.
- Reduced average repair time by 30 minutes by authoring comprehensive technical manuals for switch configuration and fiber-optic fusion, improving team-wide operational consistency.

PROJECTS

STM32-based Robot Actuator Controller (Personal Project) September 2025 - Present

- Developed an STM32-based actuator controller with UART command interface, PWM motor control, and encoder feedback
- Transitioned from time-based (open-loop) to encoder-based closed-loop position control, reducing motion variation (~17%) to $\pm 2\%$ and improving positioning consistency by $\sim 5\times$
- Implemented home referencing and software limits to ensure safe and repeatable operation

Motor Driver PCB Design (DRV8833) (Personal Project) March 2026

- Designed a 2-layer DRV8833 motor driver PCB in Altium for STM32-based actuator systems with separate 6V motor and 3.3V logic domains
- Engineered power integrity through bulk and local decoupling capacitor placement, ground pours, and via stitching to stabilize motor operation
- Optimized PCB layout based on current paths and motor power requirements, ensuring reliable bidirectional control and clean signal routing

Thermal Sensor PCB (Mars Rover) September 2025 - December 2025

- Designed and fabricated a PCB integrating TMP1075 temperature sensor with STM32 via I2C interface
- Implemented stable power regulation (5V $\rightarrow$ 3.3V) and pull-up resistor network to ensure reliable I2C communication
- Validated system through hardware bring-up and confirmed stable sensor readings under integrated system conditions

Interactive Embedded System – Lego City February 2023 - April 2023

- Built an interactive system integrating sensors, mechanical components, and Arduino-based control
- Developed MATLAB-based visualization to display real-time system behavior Designed and fabricated a PCB integrating